Silverline House — 25-Storey Residential HRB

Fire Strategy Report

Part B — Fire Safety | AD B / BS 9999 / BS 9991

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1 Introduction

This Fire Strategy Report has been prepared to accompany the Gateway 2 application for Building Control Approval for Silverline House, a 25-storey residential higher- risk building. The strategy has been developed in accordance with Approved Document B (Fire Safety) Volumes 1 and 2, BS 9999:2017, and BS 9991:2015. The report identifies and justifies all elements of the fire safety design including means of escape, passive fire protection, active fire suppression, fire detection, and fire service access. The strategy is supported by evacuation simulations carried out in accordance with PD 7974-6 and confirmatory computational fluid dynamics (CFD) modelling of the smoke extraction system.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Building Description

Silverline House comprises a 25-storey residential tower with 312 apartments, a ground-floor retail unit of 450 m^2, and four basement levels providing 180 car parking spaces and building services plant. The building has an overall height of 79.5 m from ground level to the parapet. The structural system is a reinforced concrete frame with flat slab floors. The external envelope consists of a ventilated rainscreen cladding system with mineral wool insulation and a non-combustible cladding material achieving A2-s1,d0 classification in accordance with BS EN 13501-1. The building is served by two protected staircases and two firefighting lifts.

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings.

3 Evacuation Strategy

The evacuation strategy for Silverline House adopts a simultaneous evacuation policy for the residential floors. Each apartment has a front door opening onto a protected corridor with a maximum travel distance to the nearest staircase of 7.5 m. The two protected staircases are located at opposite ends of the building core, more than 15 m apart in plan, ensuring they cannot be simultaneously compromised by a single fire event. Evacuation timing analysis carried out in accordance with PD 7974-6 demonstrates that all occupants can reach a place of relative safety within 3.5 minutes of alarm activation, including 30 seconds pre-travel time. Refuges are provided at each staircase on every residential floor for persons who are unable to use the stairs independently. The premises information box is located in the ground- floor lobby adjacent to the firefighting lift lobby.

Cavity barriers are to be provided throughout the facade construction in compliance with Approved Document B. Full details of cavity barrier specification and installation will be confirmed in the Fire Compartmentation Drawings.

4 Fire Compartmentation

The building is divided into fire compartments in accordance with Approved Document B Volume 1, Table B1. Each apartment forms a separate fire compartment. The protected corridors are enclosed by fire-resisting construction providing at least 30 minutes fire resistance. The staircase enclosures provide 60 minutes fire resistance. Cavity barriers are provided within the external wall construction in accordance with the requirements of Approved Document B to limit the spread of fire through concealed spaces. The cavity barrier specification and extent of installation will be confirmed in the Fire Compartmentation Drawings. Service penetrations through compartment walls and floors are sealed with intumescent collars or fire-stopping products with third-party certification to the required fire resistance period.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

4.1 Compartment Walls

Compartment walls between apartments are constructed in 200 mm dense aggregate blockwork or equivalent lightweight block achieving 60 minutes fire resistance in accordance with BS EN 1996-1-2 and the tabulated data in Approved Document B. Where the compartment wall coincides with the structural frame, the wall abuts the underside of the floor slab above and is sealed to prevent smoke passage. Junctions between compartment walls and the external facade are detailed to maintain compartmentation integrity at the facade interface. Product certificates and fire test evidence for all specified masonry units are included in the Construction Product Schedule accompanying this application.

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4.2 Compartment Floors and Ceiling Assemblies

Compartment floors are formed by the reinforced concrete flat slab construction (minimum 200 mm depth), which inherently provides 120 minutes fire resistance when designed in accordance with BS EN 1992-1-2. Where a suspended ceiling is provided below the structural slab, the ceiling system has been designed to maintain the compartment floor fire resistance rating. Penetrations through compartment floors for services (drainage, ventilation ducts, electrical conduits) are sealed with proprietary fire-stopping systems carrying third-party certification to BS EN 1366-3. The specification for these systems is contained in the Services Fire Stopping Schedule.

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5 Fire Detection and Alarm System

An automatic fire detection and alarm system has been designed in accordance with BS 5839-1 Category L2 for the common areas and Category LD2 for the individual apartments. Optical smoke detectors are installed in all protected corridors, lobbies, and staircase enclosures. Combination heat/smoke detectors are installed in apartment entrance halls. The system is connected to a central addressable control panel located in the ground-floor security room with a sub-panel in the Level 13 plant room. Manual call points are installed at each staircase landing on every floor and at each exit door at ground level. The detection system is interfaced with the building management system (BMS) to allow integration with smoke control, lift recall, and door hold-open release functions.

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5.1 Automatic Fire Detection

Apartment detection comprises addressable multi-sensor detectors (optical/heat) in the entrance hall, interconnected with a standalone sounder within the apartment to provide early warning to occupants. The detection system is designed to initiate the common area alarm on detection of fire within any apartment, triggering simultaneous evacuation. Detectors are positioned at 150 mm from the ceiling surface in accordance with BS 5839-1 Table B1. All detectors are accessible from floor level using a standard detector exchange tool without requiring scaffolding or elevated access equipment. The detector specification is contained in the Electrical Services Specification document.

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5.2 Fire Alarm System

The fire alarm system is configured to provide a single-stage alarm throughout the building on detection of fire. The alarm signal is transmitted to the fire service via an automatic alarm transmission system (ATS) with a dedicated monitored line in accordance with BS 5839-1 clause 25.2. The alarm system is supplied from the building's UPS system providing a minimum 72-hour standby capacity. The main control panel provides a graphical display of the building floor plans indicating the location of activated detectors to assist the fire service on arrival. A cause-and-effect matrix is included in Appendix C of this report, showing the response of all interfaced systems (smoke control, lifts, access control, hold-open devices) on activation of each detector zone.

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6 Active Fire Suppression

Active fire suppression systems installed in the building comprise a wet pipe sprinkler system covering all residential floors, plant rooms, and car parking basement levels; dry riser inlets at the external face of the building for fire service use; and a wet riser system with outlets on each residential floor. All suppression systems have been designed by a specialist MEP engineer with LPCB certification and comply with the relevant BS EN and Loss Prevention Standards. Design calculations and system schematics are contained in the MEP Sprinkler Design Specification document.

6.2 Wet Riser System

Dry risers are provided in each staircase enclosure with inlets at ground level within 18 m of a suitable fire appliance access point, and outlets at each floor level in the staircase lobby. The riser is sized to BS 9990:2015 for a building height exceeding 18 m. Wet risers are provided in each firefighting lift lobby with a dedicated pumping arrangement and a 45,000 litre break tank located in the basement plant room. The wet riser system is designed to deliver 1,500 l/min at 4.5 bar at the highest outlet simultaneously from two hose reels, in accordance with BS 9990:2015. The water supply arrangement and pump specification are detailed in the MEP Sprinkler Design Specification.

6.3 Sprinkler System

A wet pipe sprinkler system complying with BS EN 12845:2015 is installed throughout all residential floors, the retail unit, all basement car parking levels, and plant rooms. The system is classified as Ordinary Hazard Group 2 for the residential areas and Extra Hazard Group 1 for the basement car park. Design density and area of operation comply with BS EN 12845 Table 5.

The sprinkler system is designed using standard response heads with an activation temperature of 68°C in accordance with BS EN 12845:2015 Table 4. This activation temperature is appropriate for the ambient conditions in the residential areas and plant rooms. The sprinkler heads comply with BS EN 12259-1 and are from an LPCB-certificated manufacturer.

The water supply to the sprinkler system is from a 100,000 litre break tank in the basement plant room, pressurised by a duty-and-standby pump set. The pump arrangement provides a minimum flow rate of 1,800 l/min at 3.5 bar at the highest sprinkler head. The full sprinkler system design, including hydraulic calculations, is contained in the MEP Sprinkler Design Specification (document ref 2.2.3).

7 Fire Service Access

Fire service access is provided via a perimeter access road capable of supporting a fire appliance with a maximum axle load of 16.3 tonnes. The access road provides a clear width of 3.7 m and a clear height of 3.7 m under all overhead obstructions. Two firefighting lifts serving all floors are provided in accordance with BS EN 81-72:2020 and Approved Document B Section 18.

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